

Leon D. Lotter

Dr. med.

✉ leondlotter@gmail.com ✉ l.lotter@fz-juelich.de

🌐 leondlotter.de 🎓 Leon D. Lotter 🦋 leondlotter.de 📄 Leon D. Lotter 🔄 LeonDLotter

Positions

Sep 2023 – Ongoing	Research associate (100%) <i>Institute of Systems Neuroscience, University Hospital Düsseldorf, Germany</i> <i>INM-7: Brain and Behaviour, Research Centre Jülich, Germany</i> • Working group: Biomarker Development (Jürgen Dukart)
Sep 2022 – Aug 2023	Doctoral fellow (stipend) <i>Clinician Scientist Program, Max Planck School of Cognition, Germany</i> • Graduate courses in neuroscience • Lab rotation: INM-1: Structural and Functional Organisation of the Brain, Research Centre Jülich (Katrin Amunts Timo Dickscheid) • Lab rotation: Max Planck Institute of Psychiatry, Munich (Elisabeth Binder Victor Spoormaker)
Apr 2022 – Aug 2022	Research assistant (65%) <i>INM-7: Brain and Behaviour, Research Centre Jülich, Germany</i> • Project: Linking cortical thickness development to multilevel brain systems
May 2019 – Jan 2022	Student research assistant (20%) <i>Child Neuropsychology, University Hospital RWTH Aachen, Germany</i> • Diverse projects involving neuroimaging, behavioral data analysis, and visualization
Sep 2013 – Aug 2014	Voluntary service <i>Samuha Samarthya, India Service Civil International, Germany</i> • <i>weltwärts</i> program, German government-funded • Creating barrier-free environments for people with disabilities in rural South India

Education

Sep 2022 – Ongoing	PhD in neuroscience <i>Clinician Scientist Program, Max Planck School of Cognition, Germany</i> • Cumulative thesis on the neurobiological basis of human brain imaging • Supervision: Jürgen Dukart Simon Eickhoff Julian Koenig Svenja Caspers • Completion planned in 2027
Dec 2017 – Sep 2022	Doctor of medicine (Dr. med.) <i>Child Neuropsychology, University Hospital RWTH Aachen, Germany</i> <i>INM-7: Brain and Behaviour, Research Centre Jülich, Germany</i> • Thesis: Recovery-associated resting-state activity and connectivity alterations in anorexia nervosa • Supervision: Kerstin Konrad Jürgen Dukart Jochen Seitz • Grade: <i>summa cum laude</i> (with distinction)
Oct 2014 – Nov 2021	Medical studies (state examination) <i>RWTH Aachen University, Germany</i> • Elective subject: Clinical neuroscience • Clinical internships: Child and adolescent psychiatry (6 mos) Adult psychiatry (1 mo) Internal medicine (4 mos) General surgery (4 mos)

Grants & Research Funding

2025	Co-investigator: DFG Research Grant • Project: Linking brain functional trajectories in neurodevelopmental disorders to underlying neurotransmitter systems • Substantial contribution to conception and writing • Direct supervision of two researchers employed on the grant, until 2028
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Awards & Scholarships

April 2025	Heine Research Academies (HeRA) travel grant of the Heinrich Heine University • To present at SOBP 2025 in Toronto, Canada
Dec 2024	DGPPN 2024 "Best Free Talks Abstract" award • See conference section, awarded with 500€
Oct 2023	"Borchers Badge" for excellent dissertations at RWTH Aachen University • For the medical doctoral thesis "Recovery-associated resting-state activity and connectivity alterations in anorexia nervosa" passed with distinction in 2022
Jul 2023	German Academic Exchange Service (DAAD) travel grant • To present at OHBM 2023 in Montréal, Canada
Jun 2023	72nd Lindau Nobel Laureate Meeting (Physiology and Medicine) Lindau, Germany • Participation as "Young Scientist", supported by Research Centre Jülich
Sep 2022	Doctoral fellowship of the Max Planck School of Cognition • Clinician Scientist Program

Supervision & Teaching

Postdoctoral	Co-supervisor: Rachel Bamberger, Research Centre Jülich (since Jan 2026; DFG grant)
Doctoral	Co-supervisor: Naimi Quitschau, University Hospital Leipzig (since Jan 2026; DFG grant)
Bachelor's	Co-supervisor: Laura Buchloh, University of Regensburg (completed 2026) Co-supervisor: Alfred Dawalibi, Aachen University of Applied Sciences (completed 2026)
Supervisor training	Workshop "Leading & Coaching Doctoral Researchers", JuDocS qualification program for supervisors, Research Centre Jülich (Mar 2026)
Teaching	Substitute lectures, Translational Neuroscience (MSc), Heinrich Heine University Düsseldorf

Academic Contributions & Personal Skills

Software Tools	NiSpace Neuroimaging Spatial Colocalization Environment mapconn Neurophysiologically Enriched Organization of Functional Connectivity (NEOFC) ABAnnotate Gene-category enrichment analysis of brain maps using Allen Human Brain Atlas expression data
Peer Reviews	<i>Journal of the American Academy of Child and Adolescent Psychiatry</i> <i>Neuropsychopharmacology</i> <i>Translational Psychiatry</i> <i>Schizophrenia Bulletin</i> <i>Neuroimage</i> <i>Neuroimage Clinical</i> <i>Cortex</i> <i>Frontiers in Human Neuroscience</i> <i>BMC Psychiatry</i> <i>BMJ Open</i> <i>European Journal of Neuroscience</i>
Extracurricular Activities	Cohort Student Representative of the Max Planck School of Cognition (since Sep 2022) Co-organizer of the INM-7 Journal Club , Research Centre Jülich (since Oct 2023)
Programming Languages	Python since 2021 (<i>example</i>) R since 2020 Matlab since 2019 (<i>example</i>) German native speaker English professional proficiency

Conference Presentations & Workshop Participation

Jun 2026	Organization for Human Brain Mapping (OHBM) Annual Meeting Bordeaux, France Poster: Linking human brain functional connectivity to underlying neurotransmitter systems
April 2025	Society of Biological Psychiatry (SOBP) Toronto, Canada Talk: Molecular imaging-informed biomarkers link typical and atypical functional brain synchronization to underlying neurotransmission
Dec 2024	Congress of the German Association for Psychiatry and Neurology (DGPPN) Berlin, Germany Talk: Evidence of dopaminergic modulation of resting state functional connectivity alterations in psychosis
Aug 2024	Cologne Summer School in Biological Psychiatry Cologne, Germany Participation in lectures and practical exercises on state-of-the-art methods in biological psychiatry
Jun 2024	Organization for Human Brain Mapping (OHBM) Annual Meeting Seoul, South Korea - Poster: NiSpace – Neuroimaging Spatial Colocalization Environment - Poster: Neurotransmitter systems explain lifespan changes of human resting-state brain activity
Jul 2023	Organization for Human Brain Mapping (OHBM) Annual Meeting Montréal, Canada Poster: Human cortex development is shaped by molecular and cellular brain systems
Mar 2023	Minerva Symposium: Interactive Brains - From Methods to Applications Tel Aviv, Israel Invited talk: Revealing the neurobiology underlying interpersonal neural synchronization with multimodal data fusion
Mar 2023	Meeting of the German Association for Child and Adolescent Psychiatry (DGKJP) Essen, Germany Poster: Linking cortical thickness development to molecular and cellular brain systems

Publications & Preprints

Preprint	Lotter, L. D., Shafiei, G., Larabi, D., Koushik, A., Dipasquale, O., Mehta, M., Cercignani, M., Sethi, A., Harrison, N., Holiga, Š., Umbricht, D., Yakushev, I., Muthukumaraswamy, S., Forsyth, A., Hipp, J. F., Misić, B., Caspers, S., Koenig, J., Patil, K. R., Paquola, C., Eickhoff, S. B., and Dukart, J. “Linking human brain functional connectivity to underlying neurotransmitter systems”. In: <i>bioRxiv</i>. Afshani, M., Saberi, A., Reimann, G. M., Chu, C., Elmenhorst, D., Lotter, L. D., Valk, S. L., Dukart, J., Genon, S., Müller, V. I., Eickhoff, S. B., and Tahmasian, M. “Convergent connectivity of sleep disorders and its neurotransmitter and cell enrichment correlates”. In: <i>medRxiv</i>. Bamberger, R., Kuhles, G., Lotter, L. D., Dukart, J., Konrad, K., Günther, T., Siniatchkin, M., Fuchs, M., and von Polier, G. “Speech-based markers in paediatric ADHD: a longitudinal case-control study of voice features and medication effects”. In: <i>medRxiv</i>.
2026	Bamberger, R., Lotter, L. D., Nieto, N., Poulain, T., Körner, A., Kiess, W., Fuchs, M., and von Polier, G. “Voices of change: Associations between vocal markers and symptoms of ADHD - Findings from the LIFE Child Study”. In: <i>European Child & Adolescent Psychiatry</i>. Dukart, J., Lotter, L. D., Paquola, C., Eickhoff, S. B., and Schilbach, L. “The many-to-many problem of endophenotypes in psychiatry - a biological perspective”. In: <i>Molecular Psychiatry</i>. Keller, L., Lotter, L. D., Eickhoff, C. R., Eickhoff, S. B., Otten, K., Herpertz-Dahlmann, B., and Seitz, J. “Structural brain alterations in anorexia nervosa: A global brain volume and anatomical likelihood estimation (ALE) meta-analysis combined with a functional decoding approach”. In: <i>NeuroImage: Clinical</i>.
2025	Lotter, L. D. and Dukart, J. “Methodological Considerations: Integrating Measures Across Assessment Modalities”. In: <i>Neurobehavioral Individual Differences: A Transdisciplinary Approach To Advancing Clinical Science</i>. Ed. by R. D. Latzman and C. J. Patrick. Springer Nature Switzerland. Saberi, A., Wischniewski, K. J., Jung, K., Lotter, L. D., ... IMAGEN-Consortium, Paus, T., Dukart, J., Bernhardt, B. C., Popovych, O. V., Eickhoff, S. B., and Valk, S. L. “Adolescent maturation of cortical excitation-inhibition ratio based on individualized biophysical network modeling”. In: <i>Science Advances</i>.

- 2024 | Dukart, J., **Lotter, L. D.**, and Eickhoff, S. B. "Moving towards precision psychiatry: the hard nut of depression". In: *Signal Transduction and Targeted Therapy*.
- Lotter, L. D.**, Saberi, A., Hansen, J. Y., Mistic, B., Paquola, C., ... IMAGEN-Consortium, Nees, F., Banaschewski, T., Eickhoff, S. B., and Dukart, J. "Regional patterns of human cortex development correlate with underlying neurobiology". In: *Nature Communications*.
- Lotter, L. D.**, Nehls, S., Losse, E., Dukart, J., and Chechko, N. "Temporal dissociation between local and global functional adaptations of the maternal brain to childbirth: A longitudinal assessment". In: *Neuropsychopharmacology*.
- Kasper, J., Caspers, S., **Lotter, L. D.**, Hoffstaedter, F., Eickhoff, S. B., and Dukart, J. "Resting state changes in aging and Parkinson's disease are shaped by underlying neurotransmission – a normative modeling study". In: *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*.
- 2023 | Schloesser*, L., **Lotter*, L. D.**, Offermann, J., Borucki, K., Biemann, R., Seitz, J., Konrad, K., and Herpertz-Dahlmann, B. "Sex-dependent clinical presentation, body image, and endocrine status in long-term remitted anorexia nervosa". In: *European Eating Disorders Review*.
- Corneille, O., Havemann, J., Henderson, E. L., IJzerman, H., Hussey, I., Orban de Xivry, J.-J., Jussim, L., Holmes, N. P., Pilacinski, A., Beffara, B., Carroll, H., Outa, N. O., Lush, P., and **Lotter, L. D.** "Beware 'persuasive communication devices' when writing and reading scientific articles". In: *eLife*.
- Lotter, L. D.**, Kohl, S. H., Gerloff, C., Bell, L., Niephaus, A., Kruppa, J. A., Dukart, J., Schulte-Rüther, M., Reindl, V., and Konrad, K. "Revealing the neurobiology underlying interpersonal neural synchronization with multimodal data fusion". In: *Neuroscience and Biobehavioral Reviews*.
- 2021 | **Lotter, L. D.**, von Polier, G., Offermann, J., Buettgen, K., Stanetzky, L., Eickhoff, S. B., Konrad, K., Seitz*, J., and Dukart*, J. "Recovery-associated resting-state activity and connectivity alterations in anorexia nervosa". In: *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*.
- 2020 | Pankert, K., Pankert, A., **Lotter, L. D.**, Herpertz-Dahlmann, B., and Konrad, K. "Autism spectrum symptoms in children with congenital blindness". In: *Zeitschrift für Kinder- und Jugendpsychiatrie und Psychotherapie*.

* Equal contributions